

Master Thesis Supervisor Meeting – Briefing Agenda

Mode-I Phase-Field Convergence, UEL Energy Formulation & Adaptive Localization

Date: Thursday, 08 October 2026, 10:00 · **Duration:** 45 Minutes

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Governing Supervisor Directive & Active Phase

"We need to have understood everything related to the first model before we increase complexity."

Meeting Agenda & Discussion Flow (45 Minutes)

1. Opening Decision: Mode-I Adaptive Localization & Dual-Reference Framework (00–08 min)

- **Focused 10-Page Report (report_main.pdf):** Demonstrates that evaluating the localized Step-2 field concentrates refinement in the horizontal ligament (64.12% corridor share, 14,483 FEs).
- **Dual-Reference Framework:** Disambiguates 15,192-FE paper-matched fixed benchmark anchor ($F_{\max} = 0.7578$ kN) from 57,929-FE spatial-fine adaptive convergence reference ($F_{\max} = 0.7416$ kN, Job 1410504).
- **Internal Spatial Convergence:** Corrected ET1 (14,483 FEs) is within +0.279% of the spatial-fine adaptive reference in $F_{\max}$ using $\sim 75\%$ fewer elements; -1.856% vs fixed benchmark is an unresolved discretization-family offset.

2. Priority 1: UEL Energy Formulation & Output Audit (08–18 min)

- Non-invasive energy exposure in `f42_mixed_uel.for`: exact bit-for-bit mechanical parity ($|\Delta F| = 0.0$ kN, source CE8D5EDC..., Job 1409734).
- Single-IP extraction rule: eliminates $4\times$ overcounting artifact on companion visualizer CPE4 elements.
- Disproven common discrete potential ($\partial^2\Pi/\partial\mathbf{u}\partial d \neq \partial^2\Pi/\partial d\partial\mathbf{u}$) in staggered solver.
- Observability boundary: within-increment Newton paths unpersisted; status maintained strictly as `GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED` ($\varepsilon_{\text{book}} = 0.76\%$ to 4.43%).

3. Priority 2: Multifaceted Mode-I Convergence Findings (18–28 min)

- Initial stiffness K_0 : completely stable across all grids ($\Delta K_0 \leq 0.076\%$, $K_0 \approx 137.95 \rightarrow 137.84$ kN/mm, `CONVERGED / STABLE`); *NSET line-limit defect closed.
- Tolerance sensitivity sweep: ET1 (14.5k, 0.7437 kN), ET2 (6.1k, 0.7564 kN), ET3 (5.2k, 0.7594 kN), ET5 (4.7k, 0.7654 kN); ET1 is preferred for process-zone fidelity.
- Full-horizon spatial-fine solve (Job 1410504, 8T SMP, $S_8 = 3.47\times$): 7,014 increments complete to $u = 10.0\ \mu\text{m}$ with 0 cutbacks (`Exit=0`).

4. Crack-Path Symmetry, Localization & Matched Contours (28–36 min)

- Crack centroid trajectory: damage-weighted offset $|y_c - 0.500\ \text{mm}| = 0.000\ \text{mm}$ (element-bounded pure Mode-I symmetry).
- Regularized localization band: full transverse width $w_{0.5} \approx 15.0\ \mu\text{m} \approx 2l_0$ ($l_0 = 7.5\ \mu\text{m}$).
- Matched contour maps at $u \in \{0.0050, 0.005857, 0.0070\}$ mm confirm identical planar propagation and full separation.

5. Thesis Integration & Gate 6C Promotion Decisions (36–45 min)

- Approval of formal Gate 6B closure and acceptance of `GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED`.
- Authorization to proceed to Gate 6C (Mode-I State-Transfer Energy Conservation).
- Reaffirmation of strict holds on Mode-II fracture solve (`Job-2_UEL.inp`), multi-crack models, and Gate 7 visualization.